

Chapter 2

2.1

(a) $P(-|C) = 1/4$, $P(+|\bar{C}) = 1/10$ are true.

(b) Sensitivity = $P(+|C) = \frac{3}{4}$, Specificity = $P(-|\bar{C}) = \frac{9}{10}$.

(c)

True Disease Status (X)	Diagnosis (Y)		Total
	+	−	
C	0.03	0.01	0.04
$\bar{C}$	0.096	0.864	0.96
Total	0.126	0.874	1.00

(d) Marginal distribution for the diagnosis: $P(+)$ = 0.126, $P(-)$ = 0.874.

$$P(C|+) = \frac{P(+|C)P(C)}{P(+|C)P(C) + P(+|\bar{C})P(\bar{C})} = \frac{\frac{3}{4} \times 0.04}{\frac{3}{4} \times 0.04 + \frac{1}{10} \times 0.96} = \frac{0.03}{0.03 + 0.096} = \frac{5}{21} = 0.238.$$

2.9

The difference of proportions for lung cancer: $0.00140 - 0.00010 = 0.0013$. The odds ratio for lung cancer: $\frac{0.0014/0.9986}{0.0001/0.9999} = 14.018$. The estimated odds of lung cancer for smokers equal 14.018 times the estimated odds for nonsmokers.

The difference of proportions for heart disease: $0.00669 - 0.00413 = 0.00256$. The odds ratio for heart disease: $\frac{0.00669/0.99331}{0.00413/0.99587} = 1.624$. The estimated odds of heart disease for smokers equal 1.624 times the estimated odds for nonsmokers.

Because $14.018 > 1.624$, lung cancer is more strongly related to cigarette smoking.

2.15

(a) **Response variable:** overall cancer and colorectal cancer. **Explanatory variable:** regularity of aspirin uses. For those taking aspirin regularly, the risk of colorectal cancers is at least 12% lower for those taking aspirin non-regularly.

(b) (i) Significant, because the CI doesn't include 1. (ii) Weak, since the upper bound of CI is close to 1.

2.17

(a) H_0 : Race and political party identification are independent. The chi-squared test statistic is $X^2 = 184.32 > \chi_{0.95,2}^2 = 5.99$, which suggests that race and political party identification are correlated.

(b)

Race	Political Party Identification		
	Democrat	Republican	Independent
White	-11.96679	12.99946	-0.53263
Black	11.96679	-12.99946	0.53263

Because the absolute values of some cells exceed 3, there may have some correlation between race and political party identification.

(c) For the first component, we compare (Democrat, Republican). The test statistic $X^2 = 185.45 > \chi_{0.95,1}^2 = 3.84$, $G^2 = 213.62 > \chi_{0.95,1}^2 = 3.84$.

For the second component, we compare (Democrat + Republican, Independent). The test statistic $X^2 = 0.2837 < \chi_{0.95,1}^2 = 3.84$, $G^2 = 0.2813 < \chi_{0.95,1}^2 = 3.84$.

The two results are quite different because there has big difference between White and Black in the relative numbers in the Democrat and Republican, while there doesn't have big difference between White and Black in the relative numbers in the (Democrat + Republican) and Independent.

2.21

(a) Since $X^2 = 11.524 < \chi_{0.95,9}^2 = 16.92$ (p - value = 0.2415), so there's no sufficient evidence to reject the claim that job satisfaction and income are independent. However, chi-squared test using X^2 requires large samples to get closer to chi-squared distribution, and it may be poor when some expected frequencies are less than about 5.

	very dissatisfied	a little satisfied	moderately satisfied	Very satisfied
< 5	1.44	0.73	-0.16	-1.08
5 - 15	0.75	0.87	0.60	-1.77
15 - 25	-1.12	-1.52	0.22	1.51
> 25	-1.12	-0.16	-0.73	1.51

From the standardized residuals, they don't suggest any association pattern since no entry exceeds 2 or below -2.

(b) Using CMH test, the test statistic is $M^2 = 7.0449 > \chi_{0.95,1}^2 = 3.84$, and the p-value equals $0.0079 < 0.05$, which suggests strong evidence of a nonzero correlation. The reason why results differ so much from part (a) is that chi-squared test ignores the information of order in the class while CMH test takes order into account.

2.23

(a)

(i) For $H_a: \theta > 1$, the p-value is equal to 0.3808. (ii) For $H_a: \theta \neq 1$, the p-value is equal to 0.6384.

Both tests are not significant, which imply that there's no sufficient evidence to reject the claim that the odds ratio is equal to one.

(b) For the two-sided test, the mid P-value is 0.4862. The confidence interval is (0.2752, 19.1549). Compared to the p-value of Fisher's exact test (0.6384), the advantage of mid P-value is that it diminishes the conservativeness.

2.25

(a)

Victims' Race	Defendant's Race	Death Penalty		Percentage
		Yes	No	Yes
White	White	19	132	12.6
	Black	11	52	17.5
Black	White	0	9	0.0
	Black	6	97	5.8
Total	White	19	141	17.5
	Black	17	149	3.9

For which victim's race is white, $\hat{\theta}_1 = \frac{19 \times 52}{11 \times 132} = 0.68$. The sample odds for white defendants receiving the death penalty were 68% of the sample odds for black defendants.

For which victim's race is black, $\hat{\theta}_2 = \frac{0 \times 97}{6 \times 9} = 0$. because the death penalty was never given to white defendants having black victims.

(b) For the marginal odds ratio, the estimate equals $\hat{\theta} = \frac{19 \times 149}{17 \times 141} = 1.18$. The sample odds of the death penalty for white defendants were 1.18 times than for black defendants.

This data exhibits Simpson's paradox, and the reason is that there is a confounding variable "Victims' Race", which leads to the fact that the marginal association has a different direction from the conditional

association.

The number of cases for “White kills white” is the largest combination, and the number of cases for “Black kills black” is the second largest combination. However, killing white is more likely to receive death penalty, so in the marginal association, white defendants are more likely than black defendants to receive the death penalty.

2.29

Table 2.1 Cross-classification of belief in afterlife by gender.

Gender	Belief in Afterlife		Total
	Yes	No	
Females	1230	357	1587
Males	859	413	1272
Total	2089	770	2859

(a)

Posterior distribution for π_1 is $\text{Beta}(1230 + 0.5, 1587 - 1230 + 0.5) \equiv \text{Beta}(1230.5, 357.5)$.

The posterior mean estimate for π_1 is $\frac{1230.5}{1230.5+357.5} = 0.775$.

Posterior distribution for π_2 is $\text{Beta}(859 + 0.5, 1272 - 859 + 0.5) \equiv \text{Beta}(859.5, 413.5)$.

The posterior mean estimate for π_2 is $\frac{859.5}{859.5+413.5} = 0.675$.

(b)

(i) 95% posterior interval for (1) difference of proportions: (0.067, 0.133), (2) odds ratio: (1.40, 1.96).

(ii) the posterior probability that belief in an afterlife is more probable for women than for men is equal to 1.